

Ray-Yuan (Ray) Chung, MPH, RD

A human-centered AI researcher with a background in behavioral science and health informatics

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Research Interests: Human-AI Interaction, Health AI, Explainable AI, Social Computing, Behavioral Interventions

EDUCATION

University of Washington

Seattle, WA

Doctor of Philosophy (Ph.D) candidate in Biomedical and Health Informatics

2023 - 2027 (anticipated)

Concentration: Human-computer Interaction and Data Science

University of Michigan

Ann Arbor, MI

Master of Public Health (M.P.H) in Nutritional Sciences (GPA: 4.0/4.0)

2018 - 2020

National Chung-Hsing University

Taichung, Taiwan

Bachelor of Science (B.S.) in Forestry (GPA: 3.9/4.0)

2013 - 2017

Concentration: Biochemistry

RESEARCH EXPERIENCE

Human-AI Interaction Researcher

Sep 2024 - present

Unita Health

Santa Clara, CA (Remote)

- Designed an LLM-powered AI agent for gastrointestinal health, integrating RAG pipelines and few-shot prompting to enhance AI output interpretability and context awareness.
- Performed heuristic evaluations and user surveys to improve user satisfaction ratings by 30%.

Graduate Student Researcher - AI/ ML & HCI

Sep 2023 - present

University of Washington

Seattle, WA

Project: LLM-powered Chatbot for Older Adults

Adviser: Oleg Zaslavsky

- Built a human-centered chatbot using OpenAI API with iterative design principles and persona development to enhance digital literacy and information access for older adults.
- Implemented safety guardrails through context-aware prompt engineering and reinforcement learning (RLHF), reducing toxicity and hallucination in chatbot responses.
- Led a Think-Aloud usability study and post-intervention interviews, refining chatbot design for improved feasibility, acceptability, and user satisfaction.

Project: Online Health Misinformation

Adviser: Gary Hsieh

- Performed content analysis and applied NLP pipeline with Llama on 500+ YouTube videos and 1000+ comments to classify cancer misinformation and user perception

Project: Usability Evaluation of Home Spirometer

Adviser: Andrea Hartzler

- Conducted a mixed-methods study (survey and semi-structured interviews) to evaluate the feasibility and acceptability of home spirometers among cystic fibrosis patients.

Project: Online Video Content Filter

- Annotated a dataset of YouTube video screenshots and fine-tuned a multimodal deep-learning model (BERT + ViT) to flag pro-eating disorder content.

User Experience Research Intern

Jun 2024 - Sep 2024

Dexcom

San Diego, CA

- Applied the Transtheoretical Model (TTM) to design a mixed-methods survey for clustering glucose monitors (CGM) and weight management product users, resulting in persona development.
- Collaborated with PMs, AI engineers, and UX designers to translate user data into actionable insights for future personalized product development.

Research Scientist

May 2022 - Aug 2023

Impossible Foods

Redwood City, CA

- Conducted data mining and analytics of ingredient and nutrient datasets to optimize product quality, enabling the successful commercialization of 5 products in the US market.
- Partnered with cross-functional teams, including R&D, product managers, and regulatory

stakeholders to align data insights with business strategy and consumer needs.

AI Researcher

Mar 2021 - May 2021

Heali AI

Santa Monica, CA (Remote)

- Curated a large-scale medical dataset to train an AI-driven personalized nutrition model, enhancing evidence-based dietary recommendations.

Graduate Student Researcher - Behavioral Science and Public Health

Jan 2019 - Apr 2020

University of Michigan

Ann Arbor, MI

- Managed recruitment, exercise protocols, and metabolic data collection using REDCap for an RCT study on high-intensity interval training (HIIT) and weight change.

User Experience Analyst - Team Lead

Jan 2018 - Jul 2018

TaskUs/ Valve Corporation

Taipei, Taiwan

- Led data-driven UX analysis of Steam platform user trends, optimizing customer service strategy and increasing consumer satisfaction by 180% in six months.

PUBLICATIONS

- **RY. Chung**, F. Chen, Y. Jeon. Improve Technology and eHealth Literacy in a Retirement Community via The Use of LLM-Powered Chatbot.
[GSA 2024](#) - Gerontological Society of America
- F. Chen, M. Bedmutha, **RY. Chung**, J. Sabin, W. Pratt, B. Wood, N. Wibel, A. Hartzler, T. Cohen. Automated Detection of Racial Differences in Social Signals from the Content of Clinical Conversations.
[AMIA 2024](#) - American Medical Informatics Association Conference
- Reider CA, **Chung RY**, et al. Inadequacy of Immune Health Nutrients: Intakes in US Adults, the 2005-2016 NHANES.
[Nutrients 2020](#)
- Devarshi P, **Chung RY**, et al. Nutrient Gaps in US Adults by Age and Gender: Vitamin A, D, E, K, C, Magnesium, Calcium, Choline, and Dietary Fiber.
[Journal of the Academy of Nutrition and Dietetics 2020](#)

AWARDS

- Top Scholars Award, University of Washington (2023)
- Global Scholars Award, NSF International (2018-2020)
- Maxine Moore Scholarship, University of Michigan School of Public Health (2019)

RESEARCH SKILLS

- **Tools:** Python, SQL, R, PyTorch, TensorFlow, Tableau, RedCap
- **AI/ML:** Prompt Engineering, Large language models (LLMs), Machine Learning, Natural Language Processing (NLP), Software Design, Multimodal Data Analysis, Causal Inference
- **UX & HCI:** Mixed-methods Research, Usability Testing, Heuristic Evaluation,
- Participatory Design, A/B testing, Accessibility and Inclusive Design, Data Visualization
- **Behavioral Science:** Content analysis, Behavioral Assessment, Intervention Design, Psychology Theories

OTHER PROFESSIONAL EXPERIENCE

Healthcare Product Consultant

Mar 2021 - Aug 2023

So Basic Co.

Seattle, WA

- Managed ingredient research on dietary supplements to ensure regulatory compliance.

Clinical Dietitian

Jun 2021 - Apr 2022

Eating Recovery Center

Bellevue, WA

- Conducted nutritional assessments and education for adolescents and young adults with eating and mental health disorders in residential and partial hospitalization programs.

Performance Nutrition Intern

Jan 2019 - Apr 2020

University of Michigan Athletics

Ann Arbor, MI

- Supported NCAA D1 athletes with nutrition education, and body composition assessments (DXA).

Research & Development Intern

Jun 2019 - Aug 2019

Pharmavite

West Hills, CA

- Analyzed NHANES dietary data to identify nutrient gaps among the US population and drive data-informed product innovation and public health strategies.

AFFILIATED GROUPS

- The DUB (Design, Use, Build) Group at University of Washington
- Academy of Nutrition and Dietetics (AND)
- National Strength and Conditioning Association (NSCA)

LICENSURES

- Registered Dietitian, Academy of Nutrition and Dietetics (2021 - present), Registration ID: 86289001
- Certificate of Sports Nutrition Specialist, Taiwan Society for Sports Nutrition (2017 - present)